VENU GOPAL GATTINENI

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Summary

AI/ML Engineer with 6+ years of experience designing and deploying scalable machine learning solutions across fintech and digital payment platforms. Skilled in building predictive models for fraud detection, credit risk scoring, recommendation systems, and anomaly detection using Python, PyTorch, TensorFlow, Scikit-Learn, and XGBoost. Experienced in developing data pipelines with Apache Spark, Kafka, and Hive, and deploying production-grade ML systems using AWS SageMaker, Docker, Kubernetes, and MLflow. Strong background in MLOps, NLP (BERT, Transformers), and real-time model inference APIs using FastAPI, enabling data-driven decision-making for risk, product, and analytics teams.

Skills

- **Programming Languages:** Python, SQL, Bash
- **Machine Learning & AI:** Supervised Learning, Unsupervised Learning, Deep Learning, Fraud Detection, Credit Risk Modeling, Recommendation Systems, Predictive Modeling, Feature Engineering, Model Evaluation
- **Machine Learning Frameworks:** PyTorch, TensorFlow, Scikit-Learn, XGBoost
- **Natural Language Processing:** BERT, Transformer Models, Text Classification
- **Data Processing & Big Data:** Apache Spark, Apache Kafka, Apache Hive, Delta Lake, Data Pipelines, ETL Processing
- **Cloud & ML Platforms:** AWS (SageMaker, S3)
- **MLOps & Model Lifecycle:** MLflow, Model Monitoring, Model Drift Detection, CI/CD for ML
- **Backend & API Development:** FastAPI, REST APIs, Microservices
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, GitHub Actions
- **Data Analysis:** Pandas, Exploratory Data Analysis (EDA), Data Cleaning
- **Monitoring & Observability:** Prometheus, Grafana
- **Explainable AI:** SHAP, LIME
- **Software Engineering:** System Design, Debugging, Performance Optimization, Git, Agile Development

Certifications

- Complete Computer Vision Bootcamp with PyTorch & TensorFlow | Udemy | 2025
- Complete Data Science, Machine Learning, DL, NLP Bootcamp | Udemy | 2025

Experience

AI/ML Engineer | PNC Financial Services | USA | Jun 2025 – Present

- Architected end-to-end fraud detection pipelines using Python, PyTorch, and AWS SageMaker, to process high-volume banking transactions in real time, improving fraud detection precision by 28% and protecting millions of customer transactions for risk and compliance teams.
- Designed transformer-based anomaly detection models using TensorFlow and Spark ML, to identify suspicious transaction behavior across multi-channel payment systems, reducing false positives by 22% for fraud investigation teams.
- Engineered real-time ML inference APIs using FastAPI, Docker, and Kubernetes, to deploy credit risk scoring models into microservices architecture, enabling sub-second predictions for lending platforms and underwriting teams.
- Implemented MLOps pipelines using MLflow, GitHub Actions, and Terraform, to automate model training, versioning, and deployment workflows, accelerating model release cycles by 35% for data science and DevOps teams.
- Developed feature engineering pipelines using Apache Spark and Delta Lake, to unify customer transaction and behavioral datasets across banking systems, improving model training efficiency for data science and analytics stakeholders.
- Collaborated with risk analysts and product managers, to productionize explainable AI models using SHAP and LIME, ensuring regulatory compliance and transparent decision-making across credit and fraud monitoring systems.

Senior Machine Learning Engineer | Capital One | Remote, USA | Aug 2022 – May 2023

- Spearheaded development of credit risk prediction models using XGBoost, Scikit-Learn, and Python, to evaluate loan approval risk across customer profiles, improving risk classification accuracy by 24% for lending operations.
- Built large-scale data pipelines using Apache Spark, Airflow, and AWS S3, to process billions of financial records for model training, reducing data preparation latency by 40% for data science teams.
- Deployed machine learning models using AWS SageMaker and Kubernetes, to support real-time credit scoring services for digital banking applications used by millions of customers.
- Optimized model monitoring and drift detection frameworks using Prometheus and Grafana, to track model performance in production, reducing performance degradation incidents by 30% for platform engineering teams.
- Introduced NLP-based document processing models using BERT and Hugging Face Transformers, to automate financial document classification and accelerate compliance workflows for risk management teams.
- Partnered with cross-functional teams including product managers and data analysts, to integrate ML services into cloud-native microservices using REST APIs and containerized infrastructure, improving scalability of AI-driven financial platforms.

Machine Learning Engineer | Paytm | Bangalore, India | Jul 2017 – Aug 2022

- Developed recommendation systems using collaborative filtering and deep learning models in TensorFlow, to personalize digital wallet and payment experiences, increasing user engagement by 18% across Paytm platform users.
- Implemented fraud detection models using logistic regression, random forest, and gradient boosting, to identify suspicious payment activities across millions of daily transactions, reducing fraud losses by 20% for payment security teams.
- Built data processing pipelines using Apache Kafka, Spark, and Hive, to ingest and transform real-time transaction data streams for ML model training and analytics.
- Designed customer segmentation models using clustering algorithms (K-Means, DBSCAN), to support targeted marketing campaigns and improve campaign conversion rates by 15% for growth teams.
- Containerized machine learning services using Docker and deployed them on Kubernetes, enabling scalable model inference across Paytm's high-traffic payment ecosystem.
- Collaborated with data engineers and product teams, to integrate predictive analytics models into payment and fintech applications using Python APIs and cloud infrastructure, improving platform intelligence for business stakeholders.

Software Engineering Intern | Cashify | Remote, India | Dec 2016 – Jun 2017

- Built data preprocessing scripts using Python and Pandas, to clean and structure device resale data used for pricing prediction models supporting the refurbished device marketplace.
- Assisted in developing machine learning prototypes using Scikit-Learn, to estimate smartphone resale prices based on device attributes and historical transaction data.
- Designed RESTful backend services using Node.js and Express, to integrate ML predictions into the internal pricing tools used by product and operations teams.
- Conducted exploratory data analysis and feature engineering, to improve dataset quality and support model experimentation for pricing analytics projects.
- Automated data ingestion pipelines using SQL and Python scripts, to streamline data collection from internal databases for analytics and model training.
- Collaborated with software engineers and data analysts, to test model APIs and deploy prototypes supporting device valuation workflows.

Education

Master's: Information Technology | University of North America | Fairfax, VA | 06/2025
Bachelor's: Mechanical Engineering | Tirumala Engineering College | Andhra Pradesh, India | 12/2017